

CHAPTER III.4

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- 4.1. Let $f : X \rightarrow Y$ be an affine morphism of noetherian separated schemes (II, Ex. 5.17). Show that for any quasi-coherent sheaf $\mathcal{F}$ on X , there are natural isomorphisms for all $i \geq 0$

$$H^i(X, \mathcal{F}) \cong H^i(Y, f_*\mathcal{F}).$$

[Hint: Use (II, 5.8).]

Proof. (II, 5.8) gives us that $f_*\mathcal{F}$ is quasi-coherent, since X is noetherian. Thus, it suffices by (4.4) and X, Y noetherian, separated to give an isomorphism

$$\check{H}^\bullet(\mathcal{U}, \mathcal{F}) \cong \check{H}^\bullet(\mathcal{V}, f_*\mathcal{F})$$

for some open affine covers $\mathcal{U}$ of X and $\mathcal{V}$ of Y .

Take $\mathcal{V} = \{V_i\}$ to be some open affine cover of Y . f affine gives us that each $f^{-1}(V_i)$ is affine, so we have an open affine cover $\mathcal{U} = \{f^{-1}(V_i)\}$ of X . Noting that

$$\begin{aligned} C^p(\mathcal{V}, f_*\mathcal{F}) &= \prod_{i_0 < \dots < i_p} f_*\mathcal{F}(V_{i_0, \dots, i_p}) \\ &= \prod_{i_0 < \dots < i_p} \mathcal{F}(f^{-1}(V_{i_0, \dots, i_p})) \\ &= \prod_{i_0 < \dots < i_p} \mathcal{F}(U_{i_0, \dots, i_p}) \\ &= C^p(\mathcal{U}, \mathcal{F}) \end{aligned}$$

gives us a natural isomorphism of chain complexes $C^\bullet(\mathcal{U}, \mathcal{F}) \cong C^\bullet(\mathcal{V}, f_*\mathcal{F})$, so in particular an isomorphism $\check{H}^\bullet(\mathcal{U}, \mathcal{F}) \cong \check{H}^\bullet(\mathcal{V}, f_*\mathcal{F})$ of homology. This completes the proof. $\square$

- 4.3. Let $X = \mathbb{A}_k^2 = \text{Spec } k[x, y]$, and let $U = X - \{(0, 0)\}$. Using a suitable cover of U by open affine subsets, show that $H^1(U, \mathcal{O}_U)$ is isomorphic to the k -vector space spanned by $\{x^i y^j : i, j < 0\}$. In particular, it is infinite dimensional. (Using (3.5), this provides another proof that U is not affine.)

Proof. Consider the cover $\mathcal{V} = \{V_x, V_y\}$ of U , where $V_x = D(x)$ and $V_y = D(y)$. Open immersions are separated, so X separated implies U separated. Noetherianity of X gives us that U is locally noetherian. $U = V_x \cup V_y$ is a union of finitely many affine—in particular quasi-compact—open subschemes. Thus, U is itself quasi-compact, and hence noetherian. (4.4) gives us

$$H^\bullet(U, \mathcal{O}_U) \cong \check{H}^\bullet(\mathcal{V}, \mathcal{O}_U).$$

We write

$$\begin{aligned} C^\bullet(\mathcal{V}, \mathcal{O}_U) &= [0 \rightarrow \mathcal{O}_U(V_x) \oplus \mathcal{O}_U(V_y) \xrightarrow{d} \mathcal{O}_U(V_x \cap V_y) \rightarrow 0] \\ &= [0 \rightarrow k[x^{\pm 1}, y] \oplus k[x, y^{\pm 1}] \xrightarrow{d} k[x^{\pm 1}, y^{\pm 1}] \rightarrow 0], \end{aligned}$$

where $d(f(x, y), g(x, y)) = f(x, y) - g(x, y)$. As a vector space, then, $\text{im } d$ is spanned by $\{x^i y^j : i \geq 0\} \cup \{x^i y^j : j \geq 0\}$. Then

$$\begin{aligned} H^1(U, \mathcal{O}_U) &= \text{cok } d \\ &= \frac{k[x^{\pm 1}, y^{\pm 1}]}{\text{im } d} \\ &\cong k\langle x^i y^j : i, j < 0 \rangle, \end{aligned}$$

where $k\langle X \rangle$ denotes the k -span of X . □

4.7. Let X be a subscheme of $\mathbb{P}_k^2$ defined by a single homogeneous equation $f(x_0, x_1, x_2) = 0$ of degree d . (Do not assume f is irreducible.) Assume that $(1, 0, 0) \notin X$. Then show that X can be covered by the two open affine subsets $U = X \cap \{x_1 \neq 0\}$ and $V = X \cap \{x_2 \neq 0\}$. Now, calculate the Čech complex explicitly, and thus show that

$$\begin{aligned} \dim H^0(X, \mathcal{O}_X) &= 1, \\ \dim H^1(X, \mathcal{O}_X) &= \frac{1}{2}(d-1)(d-2). \end{aligned}$$

Proof. Since $(1, 0, 0) \notin X$, we have $X = U \cup V$. Further,

$$\begin{aligned} U &= \text{Spec} \left(\frac{k[\frac{x_0}{x_1}, \frac{x_2}{x_1}]}{(f(\frac{x_0}{x_1}, 1, \frac{x_2}{x_1}))} \right), \\ V &= \text{Spec} \left(\frac{k[\frac{x_0}{x_2}, \frac{x_1}{x_2}]}{(f(\frac{x_0}{x_2}, \frac{x_1}{x_2}, 1))} \right) \end{aligned}$$

are affine. We compute

$$C^\bullet = \left[0 \rightarrow \frac{k[\frac{x_0}{x_1}, \frac{x_2}{x_1}]}{(f(\frac{x_0}{x_1}, 1, \frac{x_2}{x_1}))} \oplus \frac{k[\frac{x_0}{x_2}, \frac{x_1}{x_2}]}{(f(\frac{x_0}{x_2}, \frac{x_1}{x_2}, 1))} \xrightarrow{d} \frac{k[y_1, y_2^{\pm 1}]}{f} \right]$$

□